TAGUIG CITY UNIVERSITY

Crypto-Battle: Development of a Client-Server Gamified Learning Platform Incorporating Cipher Algorithms and Data Analytics to Enhance Cryptography Competency among 3rd-Year Computer Science Students

A Thesis Presented to the Course Specialists
of the College of Information and Communication Technology
of Taguig City University

In Partial fulfillment of the requirement for the Degree
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CHAPTER I
INTRODUCTION

As digital threats continue to evolve, the demand for cybersecurity professionals with robust cryptography skills has never been greater. Despite the critical role cryptography plays in safeguarding digital assets, many students struggle to grasp its concepts due to outdated and passive teaching methods. This disconnect not only limits their ability to respond to modern security challenges but also contributes to the prevalence of human error in cybersecurity breaches.

The rationale for this project lies in the urgent need to modernize cryptography education. By leveraging gamification—a proven strategy for increasing engagement and motivation—this research aims to transform how students learn and interact with cryptographic concepts. The proposed gamified application introduces foundational cipher techniques through interactive, competitive, and analytical features designed to foster deeper understanding and practical problem-solving abilities. Ultimately, this approach aspires to empower the next generation of professionals with the knowledge and confidence required to address real-world security threats effectively.

Project Context

In today's digital world—where everything from sending messages to making purchases heavily relies on technology—the importance of cryptography in securing information and protecting against security threats cannot be overstated. Cryptography serves as a fundamental pillar of digital trust, yet its effectiveness is compromised by the lack of adequate understanding and skills among future professionals. Current industry analyses reveal that 95% of security breaches stem from human error (Venzion DBIR, 2023), underscoring the urgent need for improved cryptography education. However, traditional teaching methods have become ineffective and inefficient in addressing students' learning needs, creating a significant gap in cryptographic knowledge. Therefore, innovative and engaging educational approaches are essential to better equip students with the skills necessary to analyze, apply, and develop cryptographic solutions to real-world security challenges.

To address the defined problem, this research proposes an alternative approach that engages students in a more enjoyable and systematic learning environment. The study introduces various cipher techniques in a gamified application by integrating Caesar Cipher, Vigenère Cipher, and Transposition Cipher. The objective is to aid students to expand their knowledge and enhance their ability to analyze and solve cryptographic problems.

The system will incorporate gamified elements such as scoring, time-based challenges, real-time player-vs-player (Client-Server) battles, single player mode, and data analytics. In doing so, students are expected to become more knowledgeable in cryptography. The anticipated benefits of this study include bridging the gap left by traditional teaching methods, improving student engagement, deepening understanding of cryptographic algorithms, and enhancing problem-solving skills applicable to real-world cybersecurity scenarios.

Purpose and Description

This study aims to develop a gamified learning platform to introduce and enhance classical cryptographic algorithms to computer science students. The primary goals are to improve knowledge retention, foster analytical problem-solving skills, and provide empirical evidence on the effectiveness of gamification in technical education.

Crypto-Battle operates as a web and mobile-accessible application that combines adaptive learning techniques with interactive cipher challenges. The platform features progressive difficulty levels, real-time feedback mechanisms, and performance analytics to personalize the learning journey. Through competitive PvP modes and single-player mode, students engage with cryptographic problem-solving in a dynamic environment that mirrors real-world cybersecurity scenarios. While the study focuses on classical ciphers for foundational learning, the system’s scalable architecture allows for future expansion into modern cryptography. By bridging the gap between theoretical knowledge and practical application, Crypto-Battle aims to cultivate a generation of computer science graduates equipped with robust cybersecurity literacy.

Objectives of the Study

General Objective

The primary objective of this study is to design, develop, and evaluate Crypto-Battle, an interactive gamified learning platform that enhances the understanding of classical cryptographic algorithms among 3rd-year computer science students through the integration of cipher challenges, gamification elements, and data analytics.

Specific Objectives

1. To design an interactive game architecture incorporating classical ciphers, gamification elements, and adaptive feedback systems.
2. To develop a web-based application with single-player and multiplayer modes, progressive difficulty, and a performance analytics dashboard.
3. To implement data collection mechanisms that track user completion and error patterns.
4. To test and evaluate Crypto-Battle: Development of a Client-Server Gamified Learning Platform Incorporating Cipher Algorithms and Data Analytics to Enhance Cryptography Competency among 3rd-Year Computer Science Students using ISO 25010 in terms of the following:

   4.1 Functional Sustainability

   4.2 Performance Efficiency

   4.3 Compatibility

   4.4 Maintainability

   4.5 Reliability

   4.6 Usability

   4.7 Portability

4.8 Security

Conceptual Paradigm

In this section, the researchers present the conceptual paradigm that underpins the development of Crypto-Battle. A conceptual paradigm is a structured representation of the key concepts, variables, relationships, and assumptions that guide a research study. It serves as a theoretical foundation and provides a clear perspective for designing the research, analyzing data, and interpreting results. By utilizing established theories and models, the paradigm helps clarify the research problem, define objectives, and inform the selection of methods and analytical approaches. This framework is essential for connecting the study's inputs, processes, outputs, and feedback mechanisms, ensuring a systematic and coherent approach to achieving the study's goals.

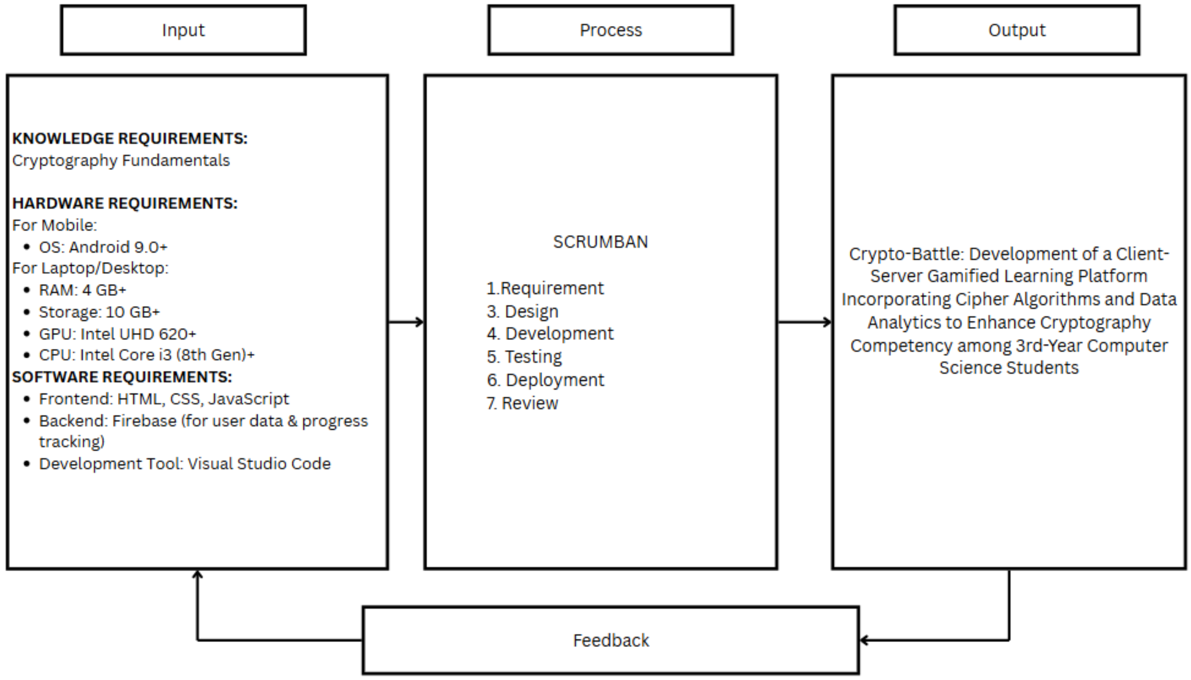

Figure 1. Conceptual Paradigm

The conceptual paradigm presented in Figure 1 illustrates the foundational framework guiding the development of Crypto-Battle. This model establishes the study's theoretical basis by defining core concepts, relationships between variables, and underlying assumptions that shape the research design. The paradigm follows an input-process-output-feedback structure, beginning with essential knowledge and technical requirements as inputs. These include fundamental cryptography concepts (Caesar, Vigenère, and Transposition ciphers) and system specifications (hardware like Android 9.0+ devices and software tools such as HTML/CSS/JavaScript and Firebase).

The process phase employs Scrumban, a hybrid agile methodology combining Scrum and Kanban, to systematically develop the game through seven stages: planning, analysis, design, implementation, testing, deployment, and maintenance. This iterative approach ensures adaptability to user needs while maintaining structured progress. The output is Crypto-Battle itself—an interactive gamified platform featuring cipher challenges, competitive PvP modes, and data analytics to track learning progress.

Finally, the feedback loop collects data from student interactions (e.g., error rates, completion times), enabling continuous refinement of the system. This cyclical process not only validates the platform's effectiveness but also informs future expansions, such as integrating modern cryptographic methods. Together, these components create a robust framework that bridges theoretical cryptography education with engaging, practical application, addressing gaps in traditional teaching methods.

Scope and Limitations

This study involves web and mobile accessibility, introducing a gamified learning platform for the third-year Bachelor of Science in Computer Science students. Introducing the three basic cryptographic techniques, including Caesar Cipher, Vigenère Cipher, and Transposition Cipher.

While the study incorporates classical cipher techniques, its applicability in advanced cybersecurity contexts is limited, as it does not cover modern cryptographic methods such as AES or RSA. Additionally, the research is confined to computer science students, which may affect the generalizability of the findings to other academic disciplines. These limitations present opportunities for future work, including expanding cipher coverage and conducting longitudinal studies to assess learning outcomes over time.

Significance of the Study

This study is significant as it contributes to cryptography education by developing an interactive and gamified learning platform tailored for third-year Bachelor of Science in Computer Science (BSCS) students. The proposed solution aims to enhance comprehension, engagement, and retention of classical cryptographic concepts, effectively addressing the limitations of traditional teaching methods.

CICT students. this research provides a transformative learning experience by converting abstract cryptographic principles into tangible, problem-solving challenges. Through real-time player-vs-player battles and progressive difficulty levels, students improve both their technical proficiency and analytical skills, better preparing them for advanced cybersecurity topics and practical applications.

CICT faculty member. will benefit from an innovative teaching supplement that presents cipher techniques in an interactive format. The platform's built-in analytics enable educators to identify common student difficulties, monitor progress, and tailor instructional strategies, thereby facilitating more personalized and effective cryptography instruction.

Aspiring cybersecurity professionals and developers. Offers foundational training in encryption logic. The scaffolded learning system lowers barriers to understanding complex algorithmic thinking, fostering essential skills for secure coding practices and threat analysis.

Educational game researchers. this project establishes a practical framework for integrating gamification mechanics-such as leaderboards, real-time feedback, and adaptive challenges-into technical education. The findings provide valuable insights that can inform future educational technology innovations in STEM fields.

Future researchers. A scalable and modular platform model that can be expanded to include modern cryptographic methods (e.g., AES, RSA) or adapted to other computer science disciplines, ensuring its ongoing relevance as cybersecurity education evolves.

Definition of Terms

Advanced Encryption Standard (AES). A symmetric block cipher that supports key lengths of 128, 192, and 256 bits, widely adopted for secure data encryption due to its strength and efficiency.

Affine Cipher. A substitution cipher that applies a linear mathematical function to encrypt characters using modular arithmetic.

Asymmetric Encryption. An encryption method using a pair of public and private keys, where one key encrypts data and the other decrypts it (e.g., RSA).

Authenticated Encryption Modes. Cryptographic methods that combine encryption and authentication to ensure both confidentiality and data integrity.

Block-Based Programming Tools. Visual programming environments (e.g., Snap!, CryptoScratch) that simplify cryptographic algorithm implementation using drag-and-drop code blocks.

Brute-Force Attack. A method of breaking ciphers by systematically trying all possible keys or combinations.

Caesar Cipher. A substitution cipher that shifts each letter in the plaintext by a fixed number of positions in the alphabet.

Confidentiality. A security principle ensuring that data is accessible only to authorized users.

CRC32 (Cyclic Redundancy Check 32-bit). A non-cryptographic hash function used for error detection in data transmission or storage.

Crypto-Battle. The interactive gamified learning platform developed in this study to teach classical ciphers (Caesar, Vigenère, Transposition) through challenges and PvP modes.

Cryptography. The practice of securing information through encryption, decryption, and related mathematical techniques.

Data Analytics. The process of analyzing user-generated data (e.g., completion rates, error patterns) to evaluate learning outcomes and system effectiveness.

Developmental Research Approach. A methodology focused on designing, developing, and evaluating educational tools (e.g., the Crypto-Battle platform).

DNA Coding (in Cryptography). An advanced technique inspired by biological DNA patterns to enhance encryption strength.

Encryption. The process of converting plaintext into ciphertext to protect data confidentiality.

Experiential Learning (Kolb's Model). A four-stage learning cycle (Concrete Experience, Reflective Observation, Abstract Conceptualization, Active Experimentation) applied in tools like CryptoEL.

Firebase. A Google-developed backend platform used in this study for user authentication, game progress tracking, and data storage.

Frequency Analysis. A cryptanalysis technique that exploits letter frequency patterns to break substitution ciphers.

Functionality (ISO/IEC 25010). Software quality characteristic assessing whether a system meets specified requirements (e.g., cipher logic accuracy).

Gamification. The integration of game elements (e.g., scoring, leaderboards) into non-game contexts to boost engagement and motivation.

Game-Based Learning (GBL). An educational approach using gameplay to teach concepts (e.g., cryptography).

Hash Function. A one-way function (e.g., SHA, MD5) that converts data into a fixed-size string for integrity verification.

Host (Multiplayer Role). A user who creates and configures game rooms in Crypto-Battle's multiplayer mode.

Integrity. A security principle ensuring data is unaltered during transmission or storage.

ISO/IEC 25010. A standard for software quality evaluation, covering functionality, usability, reliability, etc.

JavaScript. A programming language used to add interactivity (e.g., real-time cipher challenges) to the Crypto-Battle platform.

Keystream. A pseudorandom sequence used in stream ciphers to encrypt plaintext bit-by-bit.

Likert Scale. A 4-point survey scale (1=Strongly Disagree to 4=Strongly Agree) used to measure user perceptions of Crypto-Battle.

MD5. A cryptographic hash function (now considered insecure) producing a 128-bit digest.

Message Authentication Modes. To verify the authenticity and integrity of a message (e.g., HMAC).

Morse Alphabet (Morse Code). A communication method encoding characters as sequences of dots and dashes.

Phishing Detection Algorithm. Computational techniques to identify malicious phishing attempts in emails or URLs.

Player-vs-Player (PvP). A multiplayer mode where users compete in real-time cipher challenges.

Polybius Cipher. A substitution cipher replacing letters with numeric coordinates from a 5x5 grid.

Public-Key Infrastructure (PKI). A framework for managing digital certificates and asymmetric encryption keys.

Rail-Fence Cipher. A transposition cipher writing plaintext in a zigzag pattern across "rails" before reading row-wise.

Reliability (ISO/IEC 25010). Software quality metric assessing system stability and error resistance.

RSA (Rivest-Shamir-Adleman). An asymmetric encryption algorithm based on the difficulty of factoring large primes.

Scrumban. A hybrid agile methodology (Scrum + Kanban) used to develop Crypto-Battle.

Secure Hashing Algorithm (SHA). A family of cryptographic hash functions (e.g., SHA-256) for data integrity checks.

Semaphore Alphabet. A visual signaling system using flag positions to represent letters.

Stream Cipher. A symmetric cipher encrypting plaintext bit-by-bit using a keystream (e.g., RC4).

Substitution Cipher. An encryption method replacing plaintext characters with other characters (e.g., Caesar Cipher).

Symmetric Encryption. A method using the same key for encryption and decryption (e.g., AES).

Transposition Cipher. A cipher rearranging plaintext characters (e.g., Rail-Fence) without substitution.

Usability (ISO/IEC 25010). A software quality metric evaluating user-friendliness and learnability.

Vigenère Cipher. A polyalphabetic substitution cipher using a keyword to shift plaintext letters.

Visual Studio Code (VS Code). The IDE used to develop Crypto-Battle, with Firebase integration for backend services.

CHAPTER II
REVIEW OF RELATED LITERATURE

This chapter discusses the relevant literature and previous studies that helped strengthen the foundation and significance of the present research. It examines key ideas, findings, and theoretical frameworks that relate to the topic. Furthermore, this chapter includes a synthesis of these works to provide a clearer understanding of the research and to support a deeper comprehension of its purpose and direction.

Game-Based Learning for Cryptography Education

The study conducted by González-Tablas et al. (2020) introduced Crypto Go, a card game that is created to teach the foundational principles of symmetric cryptography through physical gameplay. The game is built around key learning goals that emphasizes the understanding of cryptographic basics in order to achieve the CIA triad: confidentiality, integrity and authentication. Crypto Go’s educational design utilizes deck-building mechanics and card classification to familiarize learners with cryptographic tools like stream ciphers, hash functions, message authentication codes and block ciphers.

Game-based learning (GBL) has emerged as a prospective strategy in improving engagement and learning impact when it comes to technical subjects such as cybersecurity. Based on Alghamdi and Younis (2020), games are particularly impactful in teaching practical cybersecurity concepts by enabling students to experience real-world problems in a virtual and safe setting. These games are designed with educational objectives that provide learners an environment to utilize knowledge.

To mitigate these challenges, Rahartomo et al. (2024) presented a gamified learning approach adjusted for dyslexic high school students. The study includes a workshop where 14 participants engage in a gamified and interactive activity about fundamental encryption methods. The curriculum is centered around classical ciphers such as Caesar, Playfair and Vigenère where it is presented in a level-based structured format that enables students to progress at their own pace.

According to Smith et al. (2023) CryptoGames was introduced as a gaming platform designed to teach cryptographic principles through structured gameplay. The platform consists of mini-games and mission-based challenges that reflect real-world scenarios like encrypted message exchange, brute-force attack defense, and digital signature validation. The platform has 3 progressive zones, namely Cipher Lab where classical ciphers and encryption basics are focused on, Key Arena that covers modern encryption algorithms such as RSA and Diffie-Hillman key exchange and Threat Simulator wherein learners are challenged with attack simulations that test their knowledge and strategic thinking.

In accordance to the study by Boss et al. (2023) explores how serious games can be used to teach encryption algorithms in an engaging way. By turning complex cryptographic techniques into interactive activities, students were able to better understand key concepts like substitution and transposition. The research showed that using serious games improved student participation, motivation, and comprehension compared to traditional teaching methods.

The research by Tan, M. W. Z (2023) focuses on developing a game-based platform to teach core cryptographic algorithms, including RSA and AES. The game allowed learners to interact directly with algorithm processes, helping them visualize encryption and decryption steps. Results indicated that students who used the game performed better in assessments and showed a deeper conceptual grasp of cryptography topics.

Difficulties in Learning Cryptography

Cryptographic concepts like encryption, decryption and key generation demand not only abstract reasoning but also fundamentals in mathematical literacy. As found in the study by Özer and Çetinkaya (2024), students mentioned difficulties when it comes to step-by-step processing, mathematical calculations and cognitive overload when they encounter algorithms such as Vigenère, Pigpen and Caesar ciphers. Insufficient age-appropriate guidance and alignment with cognitive development stages can greatly obstruct learning in cryptology-focused curricula.

Traditional methods such as slideshows, text-based modules and lectures have proven to be insufficient in providing users the skills to combat modern digital threats. These methods often fail to engage learners, resulting in disinterest, cognitive overload or superficial understanding. Hodhod et al (2023) emphasized that games like *CyberHero* offers an interface model that is designed to be immersive and engaging. The model is responsible for presenting information in a clear and engaging manner, making it easy for players to understand and interact with the game.

Visualization in Cryptography Education

In Visual CryptoED (2024), the authors presented a role-playing and visualization-based tool designed specifically for K-12 students. Their system allowed students to "see" the encryption and decryption processes through visual metaphors and game scenarios. They demonstrated that visualization can bridge abstract cryptographic theories into concrete, understandable concepts, especially for beginner learners.

Kothakapu et al. (2022) constructed a visualization-based cryptography module integrated in a Discrete Structure course, aimed at helping undergraduate students build their cybersecurity mindset. The plug-and-play tool, created using Unity 3D allows students to interactively explore concepts in cryptography without the need to modify the existing course structure. The module consists of three stages, first is the Basics where the fundamental cryptographic and mathematical concepts are introduced. Second is the Discussion stage where real-world applications and in-depth theoretical content are explained. Last is the Interaction stage wherein it allows learners to perform encryption and decryption tasks based on the RSA algorithm.

To address these educational challenges, Younis et al. (2020) developed the CYPHER platform that serves as an interactive learning environment designed to simplify cryptographic concepts. CYPHER integrate animated videos in illustrating real-world scenarios that explain complex protocols making them accessible to students. CYPHER's design is all about breaking down complex topics into small manageable segments allowing students to progress at their own pace. The platform's interactive nature inspires active participation, fostering a deeper understanding of cryptographic mechanisms.

Gamification in Cybersecurity and Cryptography

Gamification has become a popular strategy to increase engagement and motivation in cybersecurity and cryptography education. In Cyberverse: A Game-Based Learning Application for Cyber Security (2024), the researchers created an application where users progress through cyber defense challenges using role-play and storylines.

Gamification elements such as point systems, badges, and level progression were essential in maintaining user interest and promoting deeper learning.

Similarly, A Case Study in Gamification for a Cybersecurity Education Program (2024) emphasized the effectiveness of structured gamified activities in teaching cryptography to undergraduate students. They reported increased participation, better retention of concepts, and higher final scores in cryptography topics.

Block-Based and Interactive Programming Tools

Accessibility is another critical factor in teaching cryptography to students with diverse technical backgrounds. CryptoScratch (2023) introduced a block-based programming platform that simplified cryptographic algorithms into visual code blocks. Students without programming experience could still experiment with encryptions such as Caesar Cipher and RSA by connecting blocks, lowering the barrier to entry.

Furthermore, Rayavaram et al. (2024) introduced CryptoEL that is structured around Kolb’s Eperiential Learning. It contains four stages, namely Concrete Experience, Reflective Observation, Abstract Conceptualization and Active Experimentation. This model promotes learning by doing, thinking and applying critical components in understanding complex topics such as encryption, hashing and secure communication making the model ideal for teaching cryptography.

According to Lodi et al. (2022), a series of cryptography playgrounds was developed using a visual, block-based programming language called Snap!. These playgrounds enable students to control encryption and decryption processes directly by combining blocks that allows them to experiment on various concepts such as frequency analysis, bit-level operations and brute-force attacks.

Game-Based Learning Impact and Player Evaluations

The effectiveness of game-based learning tools must be evaluated through real-world user feedback. In Who Stole the Book of Kells? (2022), the developers created a narrative-driven cryptography game for primary school students and conducted player evaluations. Results showed not only improved conceptual understanding but also increased student enthusiasm for cryptography. The study emphasized the importance of incorporating fun, mystery, and discovery elements into educational games. They also stressed the need for careful balance: games must maintain academic integrity without becoming purely entertainment.

Gamified Cybersecurity Education

Loesch et al. (2023) introduces Secure Arcade, a gamified environment where players defend against cyber threats using knowledge of cryptography and security principles. The study highlights how incorporating mini-games with cryptographic puzzles can raise cybersecurity awareness while reinforcing encryption fundamentals. The game was effective in keeping students engaged and making complex security topics easier to approach.

Advanced Cryptographic Techniques Using Game Theory and DNA Coding

This study by Suwais (2022) proposes a novel stream cipher model that combines principles of game theory with DNA coding techniques. By applying biological data patterns and decision-making strategies, the researchers created a secure and dynamic encryption method. While the system is complex, it demonstrates innovative possibilities for integrating interdisciplinary methods into cryptography education and research.

Building on the idea of DNA coding, the follow up study of Suwais (2023) developed an enhanced stream cipher using the multiplayer version of the Prisoners' Dilemma game. The model increases encryption strength and parallelization, offering stronger resistance to attacks. Although targeted more toward advanced learners and researchers, the study shows how combining gaming theory and cryptography can inspire new teaching approaches in the future.

Synthesis of the Study

The collective body of literature highlights the growing effectiveness of game-based learning when it comes to improving cryptography education. Traditional cryptographic learning methods often presents barriers in learning due to their abstract nature and mathematical complexity. To confront these, researchers explored a diverse instructional strategy that combine interactive gameplay, gamification principles and visual metaphors to explain complex concepts. Several studies introduced cryptography focused games like Crypto Go and CryptoGames that utilizes different degree of deck-building mechanics, and real-world simulations that aim to teach classical and modern cryptographic methods.

These platforms support students in experiencing cryptographic techniques varying from classical ciphers such as Caesar and Vigenère to modern ciphers like RSA and AES, refining their engagement and comprehension. A frequent theme in the literature is the utility of visualization tools in connecting cryptographic theory with practical understanding. Platforms such as Visual CryptoED, CYPHER and Unity-based modules by Kothakapu et al. (2022) simulate how visual and interactive components can help significantly for young learners and those with cognitive learning differences.

CHAPTER III

RESEARCH METHODOLOGY

In this chapter, an overview of the research methodology employed in the study is presented. This includes the selected research approach, the population, sample size, the sampling method used, the participants involved in the study, the tools used to gather data and how their reliability was ensured, and the methods used for data collection and statistical analysis. Additionally, it discusses the technology to be implemented, and the rationale behind all technical developments for the proposed system.

Research Method Used

In conducting this study, the researchers employed a descriptive and developmental strategy. To be more precise, the researchers used a descriptive survey method to determine the sentiments of information technology professionals and faculty towards Crypto-Battle: Development of a Client-Server Gamified Learning Platform Incorporating Cipher Algorithms and Data Analytics to Enhance Cryptography Competency among 3rd-Year Computer Science Students established by the researchers. Survey and Evaluation methods was utilized in the project. The system was assessed and constructed in accordance with ISO 25010, employing frequency and percentage distribution, a four-point Likert scale, and a weighted mean. Information technology practitioners and faculty were also being asked for their recommendations.

Population, Sample Size, and Sampling Technique

The study was conducted on a selected group of fifty (50) third-year Computer Science students from the College of Information and Communications Technology at Taguig City University. Convenience sampling was used to select participants, enabling efficient data collection on their understanding of cryptographic concepts post-interaction with the game. It is a non-probability sampling method where participants are selected primarily due to their availability and ease of access to the researchers (Nikolopoulou, K., 2022).

Description of Respondents

The respondents of this study were third-year Computer Science students selected from the College of Information and Communications Technology at Taguig City University. Students were chosen based on specific criteria set by the researchers. They needed to be currently enrolled in the third year of the Computer Science program and willing to participate in the study. Additionally, the participants were expected to have at least a basic understanding of encryption and decryption techniques. The study focused on a total of fifty (50) students who assessed their learning experience through the developed interactive cipher-based game.

Research Instruments

The researchers used quantitative research, specifically with the use of survey questionnaire to collect data directly from respondents through personal distribution. The questionnaire was structured based on the ISO 25010 quality characteristics, functionality, usability, reliability, performance efficiency, maintainability, portability, compatibility, and security aiming to assess various aspects of the developed interactive learning game.

The survey utilized a four-point Likert scale, which allowed respondents to express their level of agreement or disagreement with given statements, ranging from “strongly disagree” at point 1 to “strongly agree” at point 4. This structured method provided a consistent set of answer choices, helping the researchers systematically capture the respondents' perceptions regarding the interactive learning game.

Data Gathering Procedure

To gather the necessary data for this study, the questionnaires were personally distributed to the selected third-year Computer Science students from Taguig City University. The respondents were given ample time to complete the forms during a scheduled session. The data collection process was closely monitored to ensure that instructions were followed and responses were complete. The completed questionnaires were then collected immediately after the session to ensure data integrity.

Survey Questionnaire

This survey questionnaire is designed to test and evaluate the developed interactive web-based cryptography learning game. Participants are asked to rate each item using the 4-point Likert scale provided by placing a check (✔) in the column that best reflects their level of agreement. The items in the questionnaire are aligned with the ISO 25010 software quality standards, which include key characteristics such as functionality, usability, reliability, performance efficiency, maintainability, portability, compatibility, and security.

Scale	Verbal Interpretation
4	Strongly Agree
3	Agree
2	Disagree
1	Strongly Disagree

Table 1. 4-Point Likert Scale

Table 1 presents the 4-point Likert scale used in the survey questionnaire to assess the quality of the developed interactive cryptography learning game. Each number corresponds to a specific level of agreement: 4 represents "Strongly Agree," meaning the respondent fully supports or affirms the statement. 3 stands for "Agree," indicating general or moderate agreement. 2 means "Disagree," suggesting the respondent does not support the statement. Finally, 1 corresponds to "Strongly Disagree," indicating complete opposition or rejection of the statement. This structured scale helps standardize participant responses for clear and consistent analysis.

Software Evaluation Instrument of ISO 25010

(Crypto-Battle: Development of a Client-Server Gamified Learning Platform Incorporating Cipher Algorithms and Data Analytics to Enhance Cryptography Competency among 3rd-Year Computer Science Students)

Name (Optional): ____________________________ Date: ________________

Type of Respondents:________________________

Numerical Rating and Equivalent

[4] - [2] -

[3] - [1] -

Instruction: Rate the following categories by putting a check (✔) for the favored selection before the characteristics reflected in the system. Select only one per item.

1. FUNCTIONAL SUITABILITY	**Rating Scale**			
	4	**3**	**2**	**1**
1.1 Functional Completion. Degree to which the set of functions covers all the specified tasks and user objectives.				
1.2 Functional Correctness. Degree to which the system provides the correct result with the needed degree of precision.				
1.3 Functional Appropriateness. Degree to which the functions Facilitate the accomplishment of specified task and objectives.				
TOTAL:				
2. PERFORMANCE EFFICIENCY	**Rating Scale**			
	4	**3**	**2**	**1**

2.1 Time Behavior. Degree to which the response and processing times and throughput rates of the system, when performing its functions, meet requirements.				
2.2 Resource Utilization. Degree to which the amounts and types of resources used by the system, when performing its functions, meet requirements.				
2.3 Capacity. Degree to which the maximum limits of the system parameter meet requirements.				
TOTAL:				

3. COMPATIBILITY	**Rating Scale**			
	4	**3**	**2**	**1**
3.1 Co-existence. The system can perform its required functions efficiently while sharing a common environment and resources with the product without detrimental impact on any product.				
3.2 Interoperability. The system can exchange information and the user information that has been exchanged.				
TOTAL:				
4. MAINTAINABILITY	**Rating Scale**			
	4	**3**	**2**	**1**
4.1 Modularity. Degree to which the system is composed of discrete components such that a change to one component has minimal impact on other components.				

4.2 Reusability. Degree to which an asset can be used in more than one system, or in building other assets.				
4.3 Analyzability. Degree of effectiveness and efficiency with which it is possible to assess the impact on the system of an intended change to one or more of its parts, or to diagnose a system for deficiencies or causes of failures, or to identify parts to be modified.				
4.4 Modifiability. Degree to which the system can be effectively and efficiently modified without introducing defects or degrading existing system quality.				
TOTAL:				

5. RELIABILITY	**Rating Scale**			
	4	**3**	**2**	**1**
5.1 Maturity. Degree to which the system meets needs for reliability under normal operation.				
5.2 Availability. Degree to which the system is operational and accessible when required for use.				
5.3 Fault Tolerance. Degree to which the system operates as intended despite the presence of hardware or software faults.				
5.4 Recoverability. Degree to which, in the event of an interruption or a failure, the system can recover the data directly affected and re-establish the desired state of the system.				
TOTAL:				

6. USABILITY	Rating Scale			
	4	3	2	1
6.1 Appropriateness Recognizability. Degree to which users can recognize whether the system is appropriate for their needs.				
6.2 Learnability. Degree to which the system can be used by specified users to achieve specified goals of learning to use the system with effectiveness, efficiency, freedom from risk and satisfaction in a specified context of use.				
6.3 Operability. Degree to which the system has attributes that make it easy to operate and control.				
6.4 User-Error Protection. Degree to which the system users against making errors.				
6.5 User Interface Aesthetics. Degree to which a user interface enables pleasing and satisfying interaction for the user.				
TOTAL:				
7. PORTABILITY	**Rating Scale**			
	4	**3**	**2**	**1**
7.1 Adaptability. Degree to which the system can effectively and efficiently be adapted for different or evolving hardware, software or other operational or usage environments.				
7.2 Installability. Degree of effectiveness and efficiency with which the system can be successfully installed and/or uninstalled in a specified environment.				
7.3 Replicability. Degree to which the system can replace another specified software product for the same purpose in the same environment				

TOTAL:				
8. SECURITY	**Rating Scale**			
	4	**3**	**2**	**1**
8.1 Confidentiality. Degree to which the system ensures that data are accessible only to those authorized to have access.				
8.2 Integrity. Degree to which the system prevents unauthorized access to, or modification of, computer programs or data.				
8.3 non-repudiation. Degree to which actions or events can be proven to have taken place, so that the events or actions cannot be repudiated later.				
8.4 Accountability. Degree to which the actions of an entity can be traced uniquely to the entity.				
8.5 Authenticity. Degree to which the identity of a subject or resource can be proved to be the one claimed.				
TOTAL:				

Comments/Suggestions:

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Signature

Data Analysis and Procedure

After collecting the completed survey questionnaires, the researchers organized and encoded the responses into a structured format. Each response was systematically recorded to ensure accuracy and consistency. Descriptive statistics were used to analyze the data. These tools helped interpret how participants perceived the interactive learning game based on predefined criteria aligned with ISO 25010 software quality characteristics. The analysis provided insights into areas of strength and potential improvement within the system.

Validation and Distribution of the Instrument

To ensure the reliability and accuracy of the survey instrument, the researchers submitted the questionnaire for expert validation by instructors experienced in software development. Feedback from these professionals was carefully considered to make necessary revisions that improved the clarity and relevance of the questions. Once finalized, the questionnaire was personally distributed to all the respondents for completion as part of the survey process.

Data Encoding and Formulation of the Solution

In this study, the collected responses from survey questionnaires were carefully encoded into a structured format using spreadsheet software for easy handling and organization. Each response was assigned a corresponding numerical value based on a Likert scale to ensure consistency during analysis. Once encoded, the data was processed to compute relevant statistical values.

Evaluation of Data and Result

Quantitative data were statistically analyzed using descriptive methods such as percentage and weighted mean. Tabular forms were used to assist in data interpretation by organizing the collected data into structured tables for clearer analysis. The evaluation of the gathered data focused on determining the effectiveness of the interactive game in improving students' understanding of cipher techniques, enhancing student engagement levels, and ensuring usability and functionality based on ISO 25010 standards. Based on the analysis, recommendations were developed to enhance the educational game and further support learning outcomes.

Statistical Treatment of Data

To analyze the results obtained from the survey questionnaire regarding the evaluation of the interactive learning game, as assessed by third-year Computer Science students from the College of Information and Communications Technology at Taguig City University, the following formulas were used. These calculations helped inform recommendations for enhancing the game's design and effectiveness.

1. Percentage Formula:

$$P = \frac{f}{N} \times 100$$

Where:

P = Percentage

f = Number of responses in a category

N = Total number of respondents

This formula was used to determine the percentage of respondents selecting specific responses to the survey questions.

2. Weighted Mean Formula:

$$\bar{x} = \frac{\sum fx}{n}$$

Where:

x̄ = Weighted mean

$\sum$ = Summation

f = Frequency (number of responses)

x = Weight (Likert scale value)

n = Total number of respondents

This formula was applied to calculate the weighted mean for each quality characteristic based on ISO 25010 standards, including functionality, usability, reliability, performance efficiency, maintainability, portability, compatibility, and security, as perceived by the students evaluating the interactive learning game.

Technical Requirements

This section outlines the hardware and software specifications necessary for the development and use of the web-based cryptography game.

A. Hardware Requirements

Figure 2. Laptop

The researchers used laptops with minimum specifications suitable for web-based development: an Intel Core i3 (8th Gen) processor for server operations, 4GB RAM for multitasking browsers and tools, 10GB storage for development files, and integrated graphics (Intel UHD) sufficient for web rendering. These specifications were selected to provide reliable performance while maintaining cost-effectiveness for the development environment.

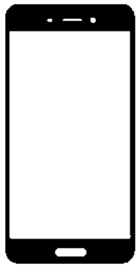

Figure 3. Mobile Phone

The system's mobile responsiveness was evaluated using Android devices, allowing researchers to assess its performance and display when accessed through smartphones. For testing purposes, the Android devices were required to run on version 9.0 (Pie) or later of the operating system.

B. Software Requirements

The figures below present the software requirements used by the researchers in the development of the proposed interactive cryptography learning game. These specifications outline the essential tools, platforms, and technologies necessary to build, test, and deploy the web-based educational system effectively.

Figure 4. Html Logo

HTML (HyperText Markup Language) was used by the researchers as the foundational language for structuring the content of the system. It provided the basic framework for building user interfaces, allowing the proper placement of elements such as buttons, input fields, game instructions, and cipher challenges. HTML ensured that all interface components were accessible and properly rendered across different web browsers.

Figure 5. CSS Logo

CSS (Cascading Style Sheets) was used by the researchers to design and style the user interface of the system. It was essential in defining the visual appearance, including layout, colors, fonts, and spacing of various game components. By using CSS, the researchers ensured that the game interface was visually appealing, consistent, and user-friendly across different devices and screen sizes.

Figure 6. JavaScript Logo

JavaScript was utilized by the researchers to add interactivity and dynamic behavior to the system. It handled essential functions such as cipher logic processing, user input validation, and real-time updates to the game interface. By using JavaScript, the game was able to respond instantly to user actions, making the learning experience more engaging and interactive.

Figure 7. Firebase Logo

Firebase is a platform developed by Google that provides backend services such as real-time databases, authentication, and cloud storage for web and mobile applications. It allows developers to build and manage apps with secure, scalable infrastructure. In this study, Firebase was used as the backend for managing username, game room, and stored performance data such as time taken per level. Its integration enabled smooth data tracking and secure access throughout the interactive cryptography learning game.

Figure 8. Visual Studio Code Logo

Visual Studio Code (VS Code) is a lightweight, open-source code editor developed by Microsoft. In this study, Visual Studio Code served as the primary integrated development environment (IDE) for building the interactive cryptography learning game. It was used to write, organize, and manage the source code efficiently. Additionally, the researchers utilized Firebase tools integrated into VS Code to enable seamless connection with Firebase services used in the game. The researchers used Visual Studio Code version 1.100.0.

Network Requirements

The network must efficiently support PC and laptop users, ensuring secure connections, reliable backup systems, and responsive performance. A minimum internet speed of 10 Mbps is recommended for optimal operation of the system, allowing smooth access to game features such as single-player mode, multiplayer mode, tutorials, techniques content loading, and real-time interactions. Speed tests will be conducted during development and deployment to ensure the system performs well under various network conditions.

The web interface will be compatible with standard browsers such as Chrome, Firefox, and Safari, ensuring a consistent experience for users. However, since the game is web-based, it requires an active internet connection to function. If there is no internet access, the game will not be operational, which underscores the importance of a stable network connection for uninterrupted usage.

Project Design

The given figures below are the initial design of the proposed system developed by the researchers. It includes a title screen, a main menu, and multiple gameplay interfaces such as single-player and multiplayer modes.

Figure 9. Game Title Screen

Figure 10. Main Menu

Figures 9 and 10 show the initial user interface of the cryptography learning game. Figure 9 displays the title screen with the game name "CRYPTO BATTLE". Figure 10 follows with the main menu, which provides access to Single Player, Multiplayer, Tutorial, and Techniques sections, along with a Username field for personalization.

Figure 11. Difficulty Selection

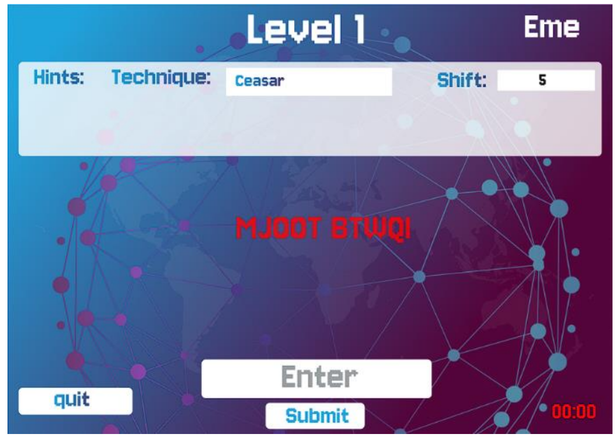

Figure 12. Easy Mode

Figures 11 and 12 present the early stages of Single Player mode. Figure 11 allows players to choose between Easy and Hard difficulty levels based on their skill or challenge preference. Figure 12 shows the Easy mode interface, which displays the cipher technique, a helpful hint, and a timer that tracks progress without pressure—ideal for beginners starting the first of 10 levels.

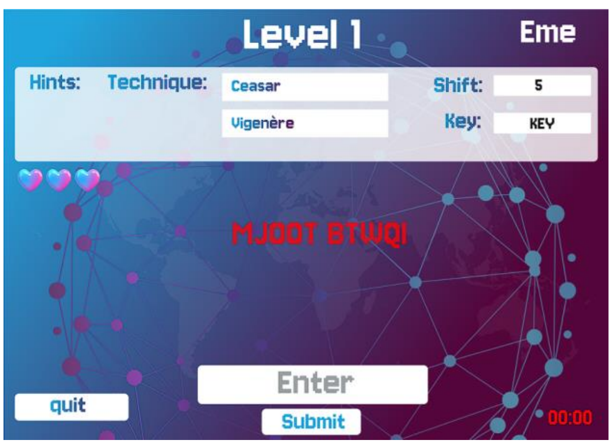

Figure 13. Hard Mode

Figure 14. Game Summary

Figures 13 and 14 show the advanced mode and performance summary. Figure 13 introduces Hard mode with multilayered ciphers, a time limit based on the cipher method used, 3 lives, and hints. Figure 14 appears after completing all 10 levels in any difficulty, displaying total time spent, and average time per level.

Figure 15. Multiplayer

Figure 16. Create Room

Figure 15 shows that upon selecting Multiplayer, users are presented with two options: Create Room or Join Room. Figure 16 displays the next step when Create Room is selected, prompting the user to choose a role: Host or Player.

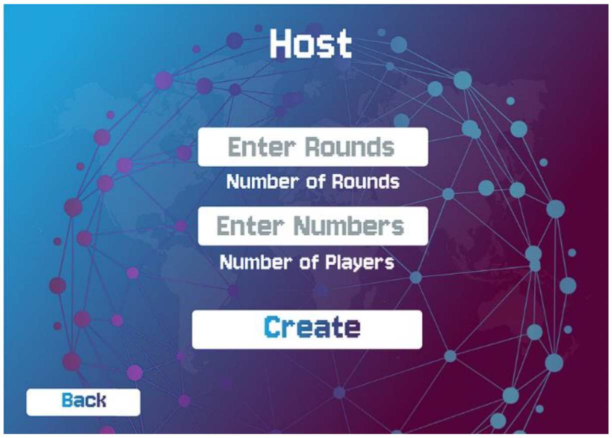

Figure 17. Host

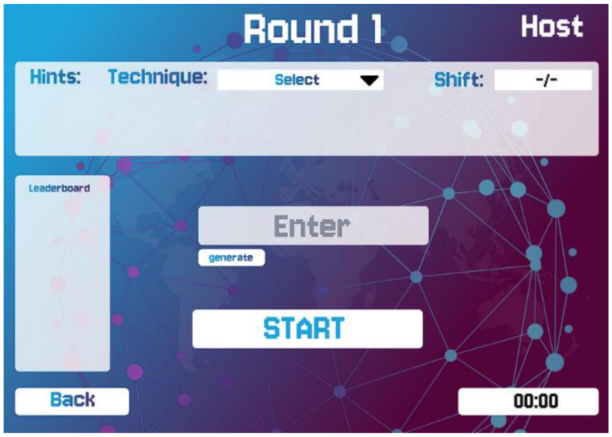

Figure 18. Configure Round (Host)

Figure 17 shows that if Host is selected, the user controls game setup, setting the number of rounds and players. The Host manages the game but does not actively play. Figure 18 illustrates that for each round, the host can input a word or use an auto-generated one, select a cipher technique, and set a custom timer. A leaderboard displays the fastest player per round.

Figure 19. Player

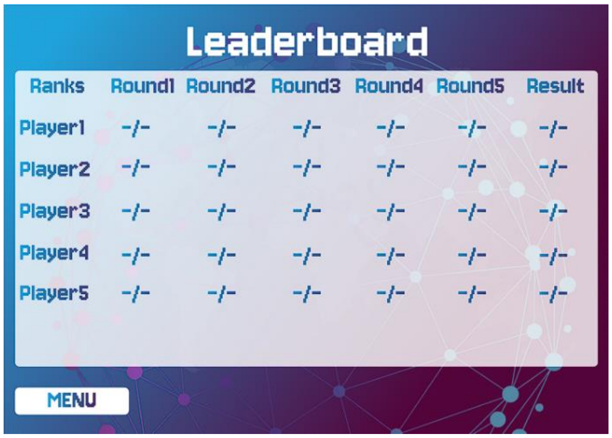

Figure 20. Final Leaderboard

Based on Figure 16, when a user selects Create Room, they choose a role: Host or Player. Figure 19 shows that if Player is selected, the user sets the number of rounds and a uniform time limit, allowing them to join the game they created. Figure 20 displays the total scores of all players across rounds in any multiplayer game, determining the final rankings at the end of each session.

Figure 21. Lobby

Based on figure 15, when a user selects Multiplayer, they are presented with two options: Create Room or Join Room. Figure 21 illustrates the scenario when the user chooses Join Room, where they can view and select from the available rooms in the lobby created by hosts and players.

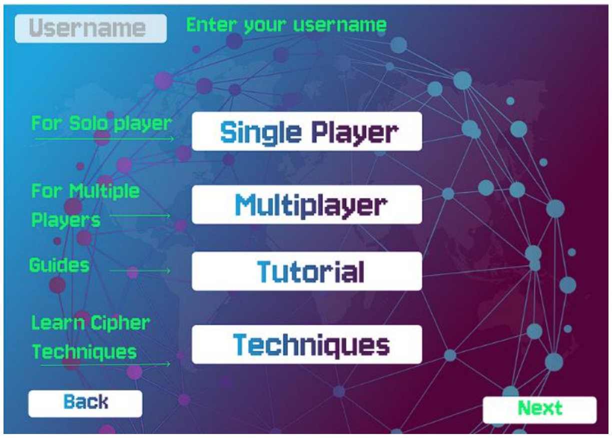

Figure 22. Tutorial **Figure 23. Techniques**

Referring to Figure 8, the main menu includes Tutorial and Techniques. Figure 22 shows the Tutorial, which explains how Crypto-Battle works. Figure 23 shows Techniques, providing an overview of Caesar, Vigenère, and Transposition ciphers to help users understand each method.

Diagrams

Diagrams are the constructed visualization of the researchers to guide them on how to create the proposed system. This part consists of the System Architecture, Data Flow Diagram, Flowchart, Unified Modeling Language, and System Development. Initial diagrams were created for each category to better understand the system's functionality.

System Architecture

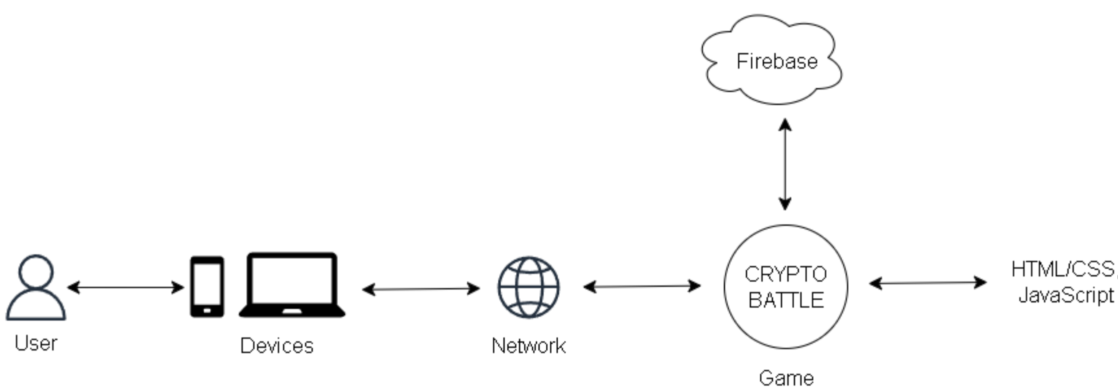

Figure 24. System Architecture of Crypto-battle

Figure 24 shows the system architecture of Crypto-Battle, illustrating how the web-based cryptography game functions. Players access the game from various devices including mobile phones or laptops through a browser. The frontend is built with HTML, CSS, and JavaScript, providing the interactive game interface that users see and interact with. This frontend communicates with Firebase services which handle the backend operations - processing game data, storing player progress, and game result in real-time. The entire system leverages Firebase's cloud infrastructure to ensure smooth performance and scalability, even during peak multiplayer sessions. This architecture is particularly effective for browser-based educational games that require real-time data synchronization and cross-platform accessibility.

Data Flow Diagram

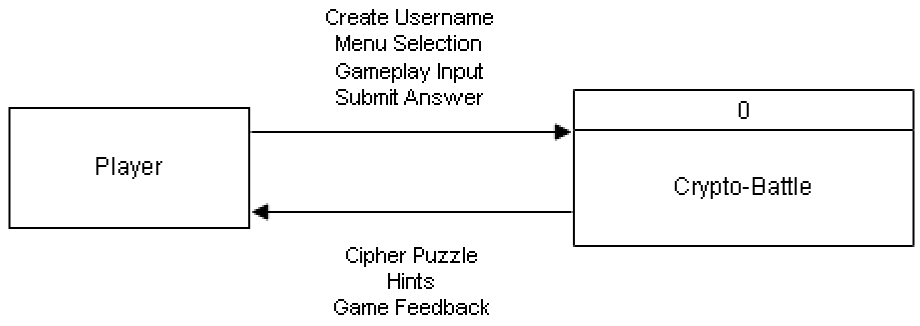

Figure 25. Data Flow Diagram Context Level 0

Figure 25 illustrates the Data Flow Diagram Level 0 of the Cipher Learning Game. It represents the system as a single main process that interacts directly with the user, identified as the Player. The player provides input such as username, menu selections, gameplay actions, and cipher answers. These inputs are sent to the game system, which processes them and responds with cipher puzzles, hints, and game feedback. This includes whether the submitted answer is correct or incorrect, along with performance results. The diagram provides a simplified overview of the main data exchange between the player and the system, covering interactions in both single player and multiplayer contexts.

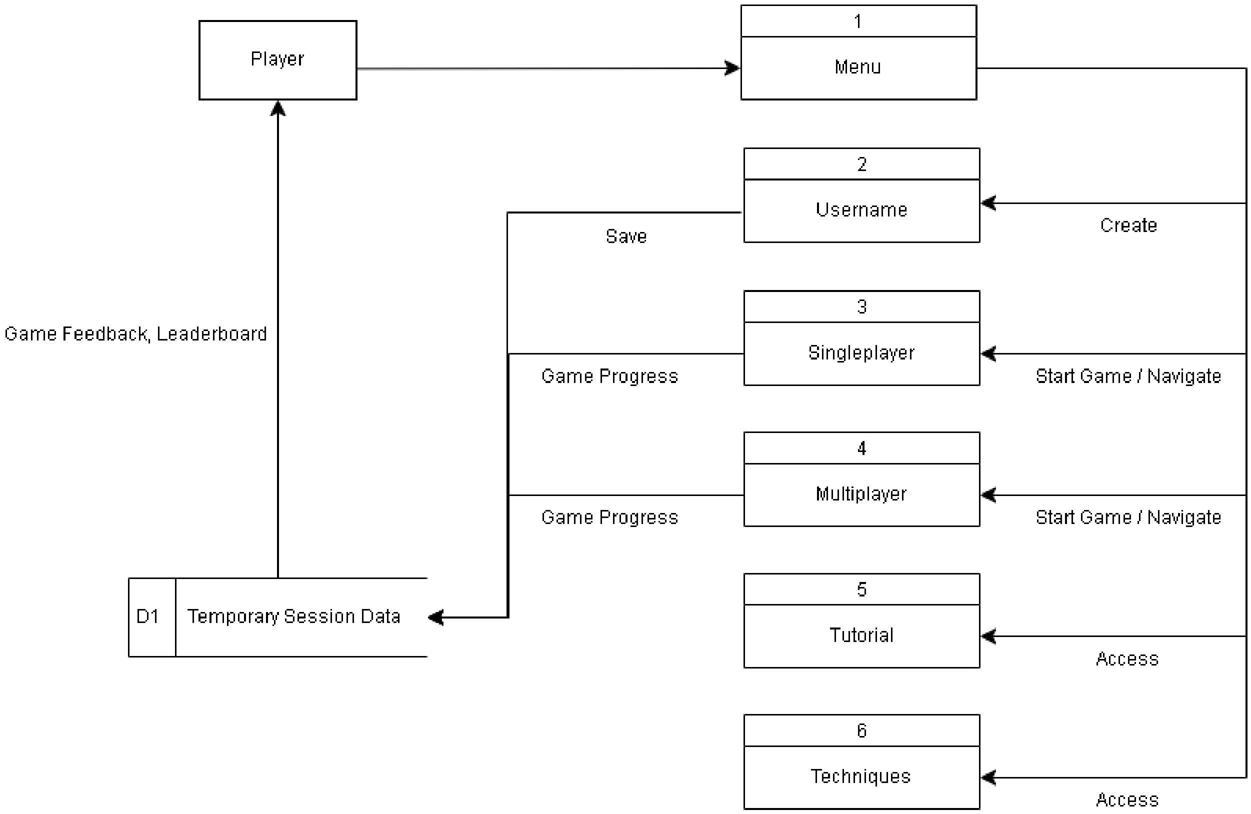

Figure 26. Data Flow Diagram Context Level 1

Figure 26 illustrates the Data Flow Diagram Level 1 of the Cipher Learning Game. This level provides a more detailed view of the internal gameplay structure. The Player interacts with the system by first entering their username which serves as the primary identifier throughout the gameplay session, then navigating the menu and selecting either Single player, Multiplayer, Tutorial, or Techniques. In Single player and Multiplayer modes, game progress is recorded, and performance is stored in a temporary session data store. The system returns game feedback and leaderboard information to the player. Tutorial and Techniques modules are accessed for guidance and learning. This diagram demonstrates how user actions flow through different system components, highlighting progress tracking and feedback delivery within the game.

Proposed Flowchart

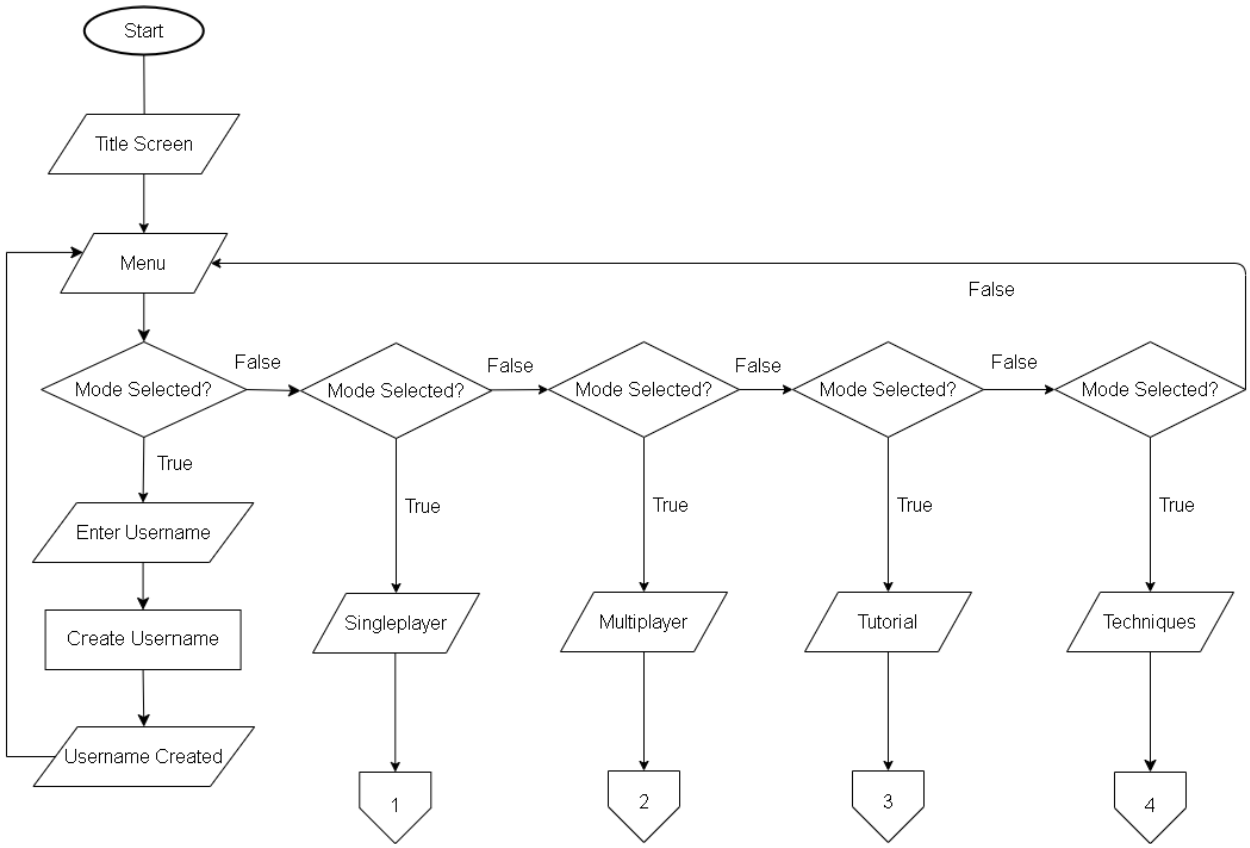

Figure 27. Game Launch Flowchart

Figure 27 illustrates the game launch process and main menu navigation. It begins with the user launching the game. The menu then displays various gameplay options such as Enter Username, Single Player, Multiplayer, Tutorial, and Techniques. The system waits for user input, and once an option is selected, it stores the choice and transitions to the corresponding section. This structured process ensures a smooth and intuitive user experience from startup to game interaction.

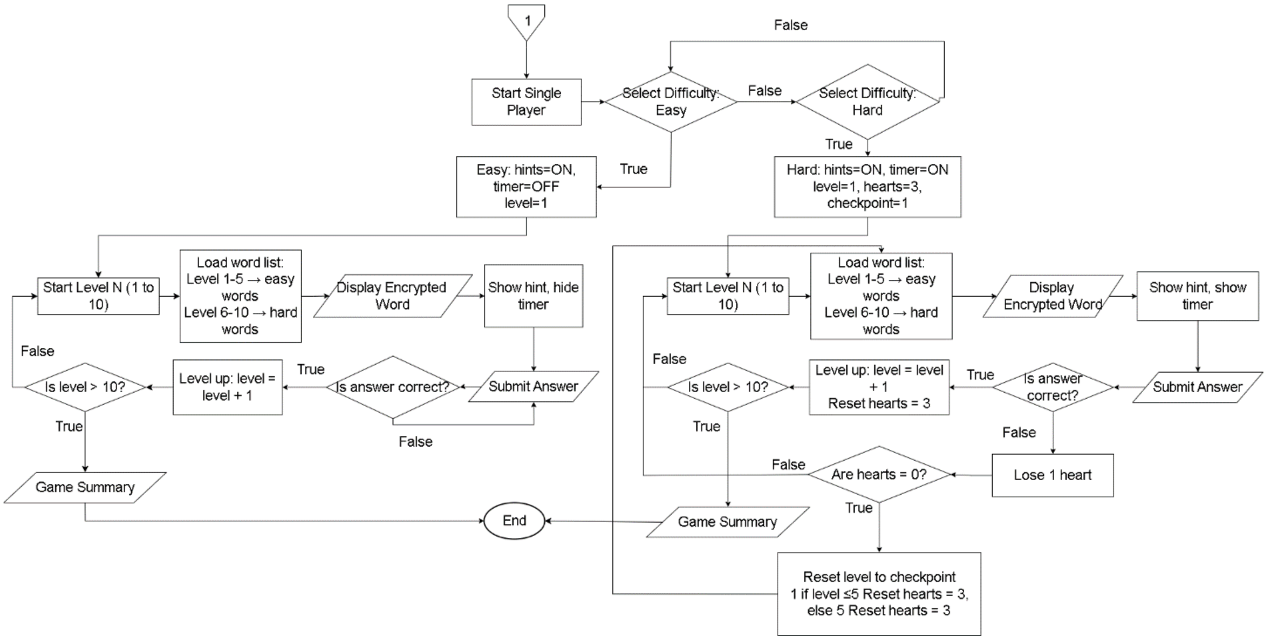

Figure 28. Single-player Flowchart

Figure 28 represents the process of a single-player game. It begins with selecting the difficulty level: "Easy" provides hints and has no timer, while "Hard" has a timer and hints. Both paths lead to a core game loop: loading a level-specific word list, displaying an encrypted word, and showing hints/timer. After submitting an answer, if correct, the player levels up and continues the loop, ending the game if Level 10 is surpassed. If incorrect, Easy mode allows retries. Hard mode deducts a heart; if hearts reach zero, the game ends. Otherwise, if hearts remain, the level resets to the checkpoint, hearts reset to 3, and the game loop continues until completion, concluding with a game summary.

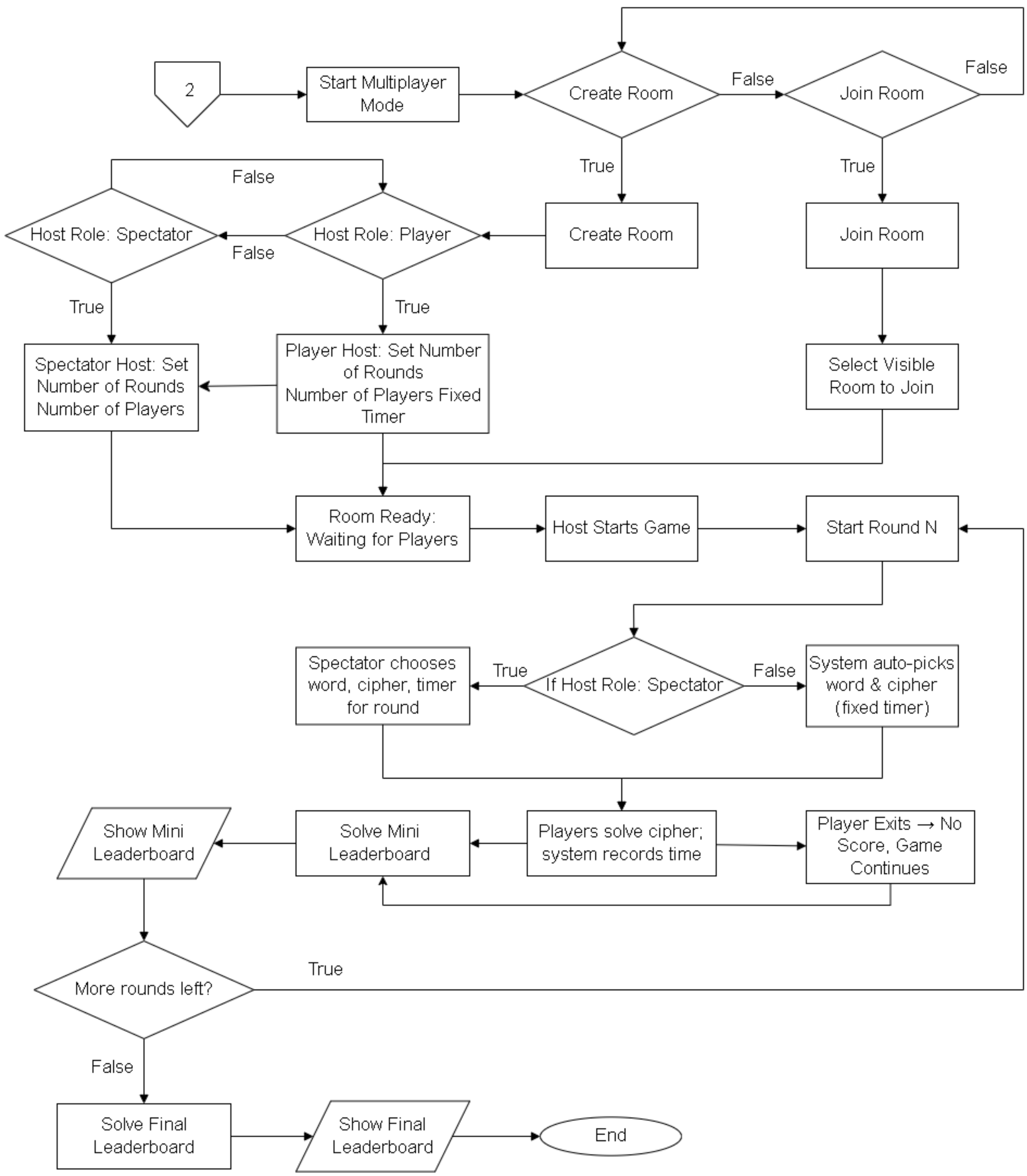

Figure 29. Multiplayer Flowchart

Figure 29 depicts the multiplayer game flow. Users initiate "Multiplayer Mode," then choose to "Create Room" or "Join Room." Creating a room involves selecting a "Host Role"

(Spectator or Player) and configuring game settings (rounds, players, timer). The room then waits for players. Joining a room involves selecting an existing room. Once the room is ready, the "Host Starts Game," initiating rounds. Spectator hosts pick the round's word/cipher/timer. Players solve the cipher, their time is recorded, and a "Mini Leaderboard" displays. Rounds continue until no more are left. Player exits result in no score, but the game continues. The process concludes with a "Final Leaderboard."

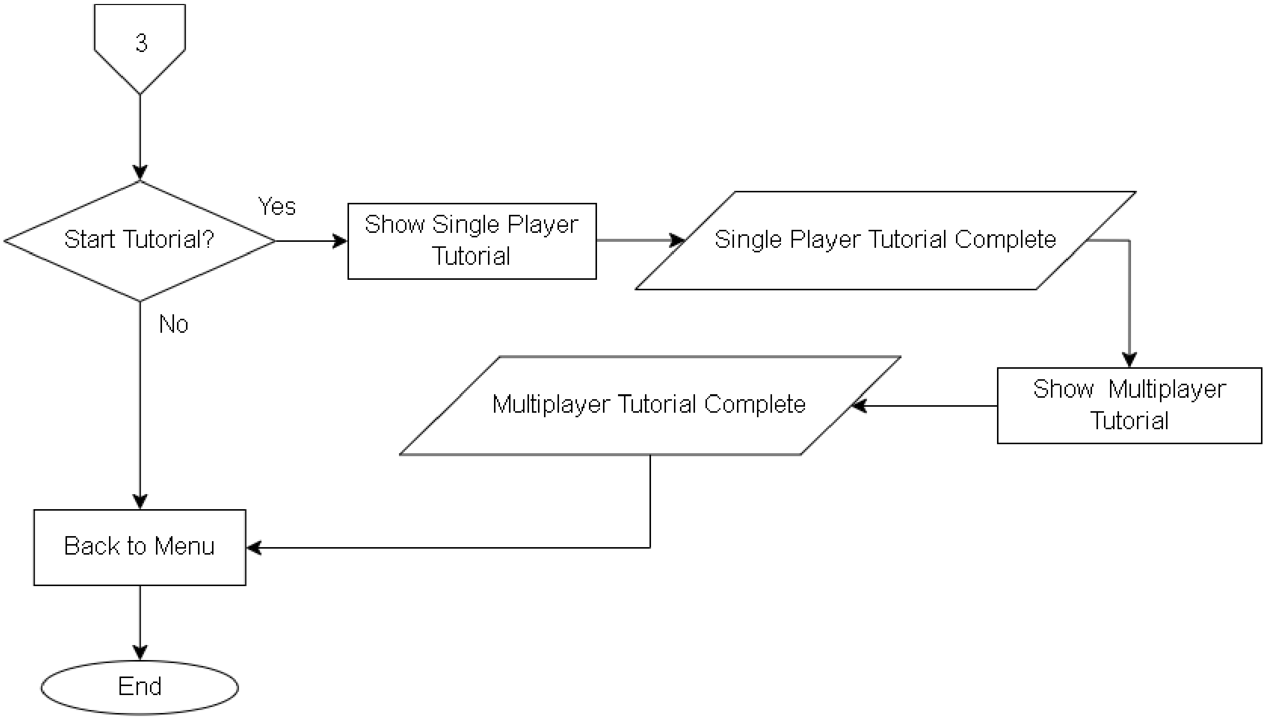

Figure 30. Tutorial Flowchart

Figure 30, the Tutorial Flowchart, outlines the user's path through the instructional system. The process begins with an inquiry about starting the tutorial. If the user declines, the sequence moves directly to the main menu and concludes. If the user agrees, they first engage with the initial tutorial content, which then proceeds to the single-player instructional segment. After the single-player tutorial concludes, the system transitions to the multiplayer instructional segment. Once the multiplayer instruction is complete, the flow returns to the main menu before concluding.

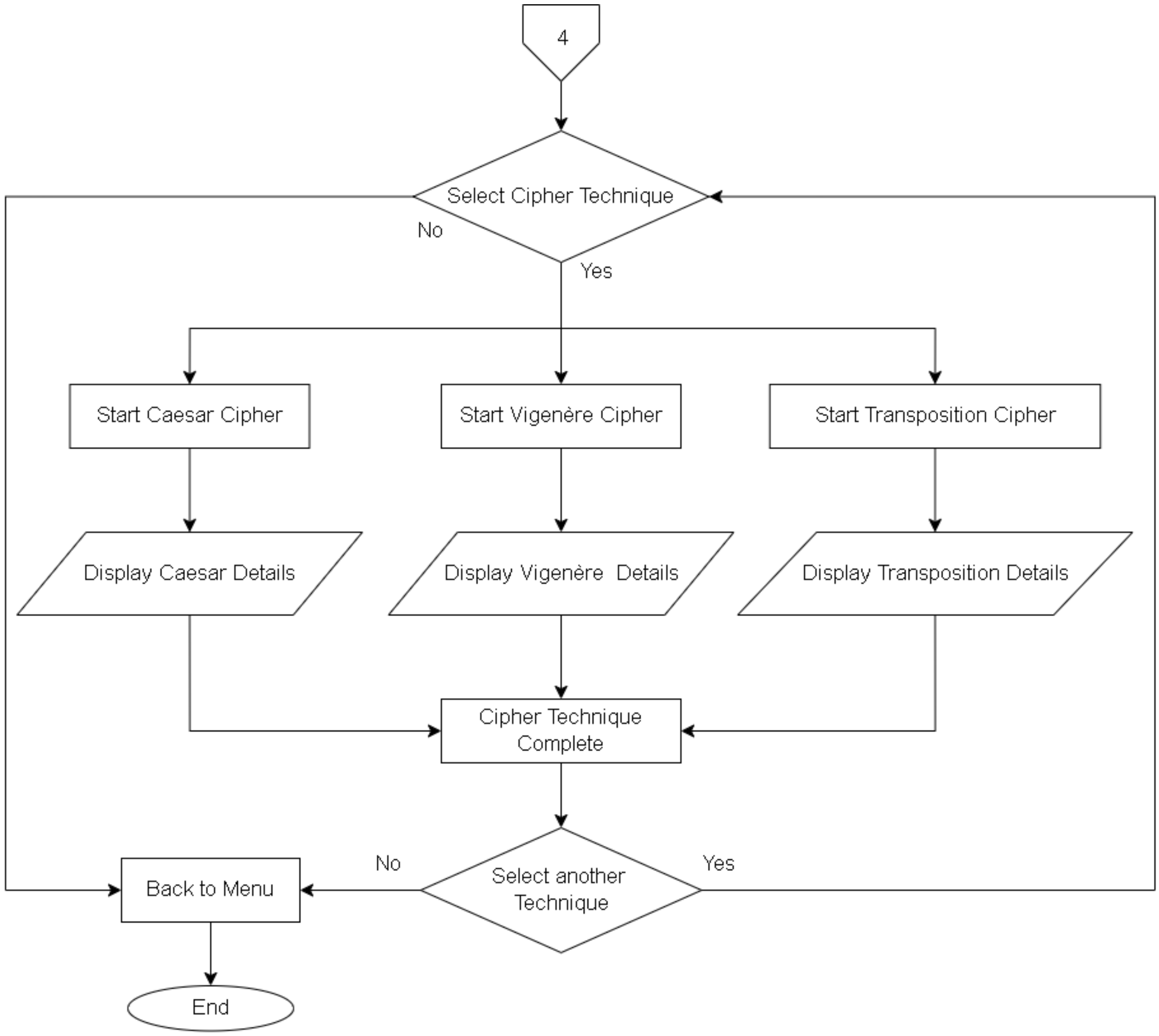

Figure 31. Techniques Flowchart

Figure 31, the Techniques Flowchart, outlines a system for learning about various cipher techniques. The user begins by selecting a specific cipher. Three distinct options are presented: Caesar Cipher, Vigenère Cipher, and Transposition Cipher. Upon choosing one, the system displays detailed information pertinent to that particular cipher. After reviewing the details, the process marks the "Cipher Technique Complete." The user then has the option to explore another technique, which loops them back to the selection point. If no further exploration is desired, the system returns to the main menu and concludes.

Unified Modeling Language

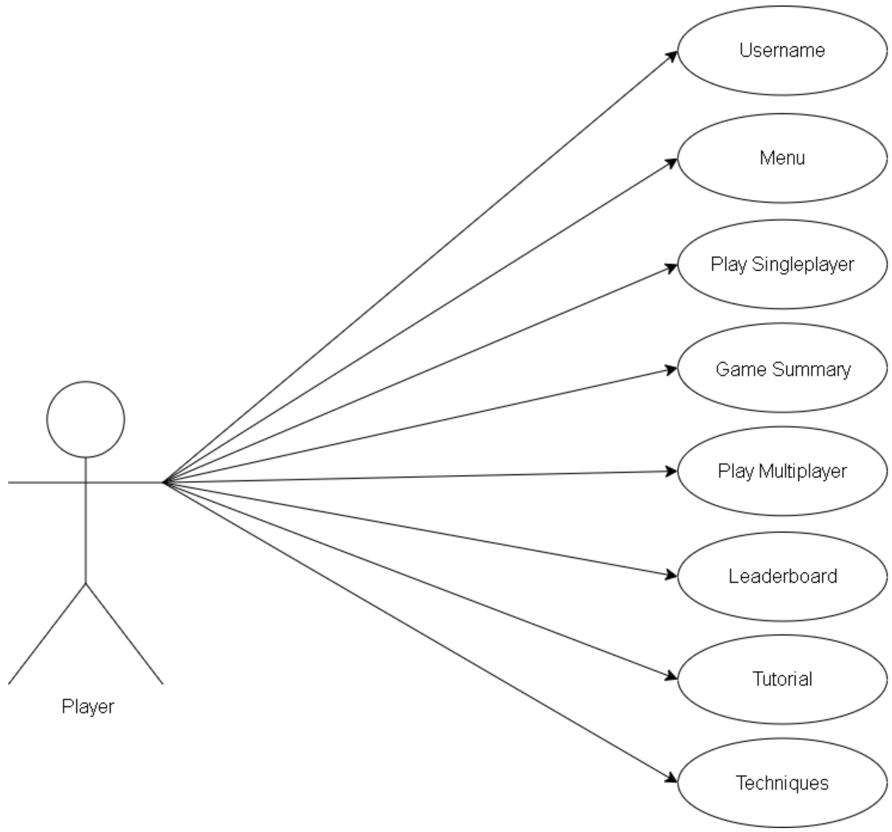

Figure 32. Use Case

Figure 32 presents the use case diagram for the Cipher Learning Game, illustrating the a "Player" actor connected to key functions: entering a username, accessing the main menu, selecting single player or multiplayer modes, viewing game summaries and leaderboards, and accessing tutorial content with technique guides. This visual representation outlines the complete player journey through the game's primary features and navigation structure.

System Development

Figure 33. Scrumban Agile Model

Figure 34 presents the system development of the system; the researchers will use the Agile Method. The researchers choose Agile Model as a methodology because the model demonstrates how the growth of the system is divided into a series of linear phases. These phases include:

Requirements

In this phase, the researchers identified the primary goals and needs of the cryptography learning game. The objective was to build an educational system that simplifies the learning of classical ciphers such as Caesar Cipher, Vigenère Cipher, and Transposition Cipher through gamified, interactive features. Consideration was given to limited resources, so lightweight, accessible technologies like HTML, CSS, JavaScript, and Firebase were selected to ensure both development feasibility and user accessibility.

Design

This phase involved planning the system's layout, user interface, and data flow. Researchers created wireframes to visualize each game mode, including Single Player and Multiplayer. The front end was designed using Visual Studio Code, with a simple, engaging color palette to attract and retain users. Navigation, tutorial integration, and real-time features were also planned to ensure a smooth learning experience.

Development

During development, the game logic, user interface, and Firebase backend were implemented. Researchers focused on key functions such as cipher-based game challenges, multiplayer interaction, and user performance tracking. The application was coded using JavaScript for game logic, with Firebase handling authentication, real-time updates, and data storage. Each feature was built incrementally to ensure flexibility and adaptation.

Testing

Testing was done continuously as new features were added. Researchers used iterative testing to catch bugs, improve usability, and verify correctness in cipher logic. The Scrumban method helped manage tasks and prioritize fixes efficiently. Feedback from test users guided UI adjustments, error corrections, and performance tuning.

Deployment

Once testing confirmed system stability, the application was deployed for actual use. Users were able to access the system online and experience the full functionality of both game modes. Researchers monitored the deployment phase closely to ensure the system performed well and handled multiple users effectively.

Review

In the final stage, the team reviewed all collected feedback and performance data. Adjustments were made based on user suggestions and observed issues. This review ensured that the system met its educational goals and provided insights for future updates or expanded versions of the cryptography learning game.

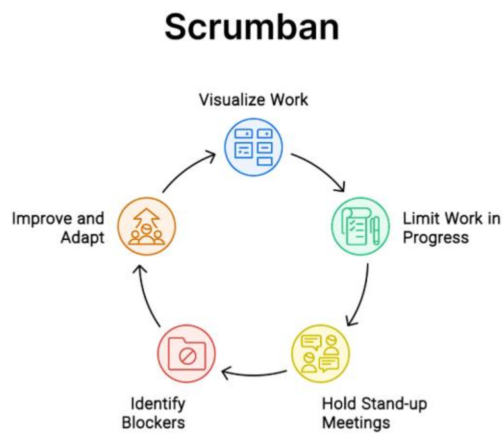

Figure 34. Scrumban Framework

Figure 35 shows the Scrumban Framework used to manage the cryptography learning game's development. Combining Scrum and Kanban, it helped the team organize tasks, limit work in progress, hold daily check-ins, and solve blockers. Below is a brief explanation of how each step was applied in the project.

Visualize Work

In the initial stage of development, the researchers applied the Scrumban framework to organize tasks clearly and visually. A digital board was created, dividing tasks into columns such as "To-Do," "In Progress," and "Done." This helped the team keep track of all activities, understand priorities, and ensure that every task required for the cryptography learning game was visible and manageable throughout the project lifecycle.

Limit Work in Progress

To maintain focus and prevent overload, the researchers limited the number of tasks being worked on at the same time. Only a few tasks were allowed in the "In Progress" column to ensure that each one received proper attention. Once a task was completed and moved to "Done," only then was a new task pulled in. This helped maintain a steady workflow and avoid confusion or burnout among the team members.

Hold Stand-up Meetings

Regular short meetings were conducted to keep the team aligned. During these stand-ups, each researcher discussed what they had completed, what they were currently working on, and any obstacles they were facing. This daily coordination ensured that everyone remained updated and any concerns or changes were addressed early, keeping the momentum of development strong.

Identify Blockers

Throughout the project, whenever the team encountered problems—such as bugs in the code, delays in feedback, or unclear instructions—these were immediately marked as blockers on the Scrumban board. Identifying these blockers helped the researchers focus on resolving critical issues quickly before proceeding to the next tasks, preventing delays in the overall timeline.

Improve and Adapt

The Scrumban framework encouraged continuous improvement. After each development cycle, the researchers reviewed what worked well and what didn't. They adjusted their workflow, tools, or task priorities to improve future output. This flexibility allowed the team to enhance their performance, fix design issues, and respond quickly to user feedback and testing results.

Algorithm Discussion

This section details the cryptographic algorithms implemented within the game, chosen specifically for their pedagogical value in demonstrating fundamental principles of classical cryptography. The game incorporates three distinct ciphers: the Caesar Cipher, Vigenère Cipher, and Transposition Cipher, each representing a unique approach to encryption and offering a progressive understanding of cryptographic complexity and security.

The Caesar Cipher is a simple substitution cipher. It works by shifting each letter a fixed number of places in the alphabet. This straightforward method clearly shows how messages can be encrypted and decrypted using a secret key. It also helps illustrate that very simple codes can be easily broken.

The Vigenère Cipher is more complex. It's a polyalphabetic substitution cipher, meaning it uses many different shifting alphabets based on a repeating keyword. This makes it much harder to crack than the Caesar Cipher using simple methods. Learning about it helps understand how using a longer, repeating key can make a code much stronger.

The Transposition Cipher uses a different approach. Instead of changing letters, it rearranges their order in the message. This is often done by putting letters into a grid and then reading them out in a different, secret pattern. This cipher teaches how just changing the position of letters can hide a message, showing another basic way to make a code.

Collectively, these three classic ciphers give a good, step-by-step way to learn about old coding methods. They effectively show the main ideas of changing letters, moving them around, managing keys, and how codes have developed to become harder to break.

Since the proposed system will implement a variety of encryption methods, the researchers have decided to perform a comparative analysis between these algorithms. This comparison seeks to evaluate the strengths, weaknesses, features, functions and uses of the selected methods. Additional encryption may be implemented in the future iterations of this study to provide a broader and more comprehensive evaluation.

This comparison intends to evaluate the efficiency, comprehensibility and conceptual clarity of the selected encryption methods in relation to teaching cryptography. By analyzing factors such as key structure and algorithmic complexity, learners can perceive how different cryptographic techniques works and why certain methods are applied in specific scenarios.

Caesar Cipher

Strengths:

- Caesar cipher is a simple and beginner-friendly encryption useful for foundational learning.
- It is also known for its fast computation and implementation making it ideal for educational purposes.

Weakness:

- It has a very weak security that can be easily broke into through brute force and frequency analysis.

Vigenère Cipher

Strengths:

- Stronger than the simple Caesar cipher.
- It also resists simple frequency analysis if the key is random and long.

Weaknesses:

- The repetition of keys makes it weak.
- It is still breakable when it comes to modern techniques.

Transposition Cipher

Strengths:

- Transposition cipher can preserve letter frequencies.
- The preservation of letter frequencies makes it more difficult to break into.

Weaknesses:

- It requires a precise key order
- Its message structure can be partially removed.

Features

Caesar Cipher

1. Monoalphabetic substitution cipher
2. Simple and fast to implement
3. Shifts each letter by a fixed number of positions according to the alphabet
4. Extremely low key space, with only 25 possible shifts
5. Mainly used for educational purposes and puzzles
6. Can be broken using brute force or frequency analysis easily

Vigenère Cipher

1. Polyalphabetic substitution cipher
2. Uses a repeating keyword to determine the shift for each letter
3. Multiple shifting alphabets makes it more secure than Caesar
4. Short or reused keys can make it vulnerable to frequency analysis
5. It has a moderate complexity and resistance to brute force attacks

Transposition Cipher

1. Rearranges the characters in the plaintext without changing them
2. Maintains letter frequency that makes it susceptible to pattern analysis
3. More secure than a simple substitution with proper key management
4. It requires more computation to encrypt and decrypt
5. Used as a foundation in more complex modern ciphers and puzzle encryption

Functions

Caesar cipher is a basic substitution cipher that replaces each letter in the plaintext with another letter a fixed number of positions down the alphabet. It is primarily a historical or instructional tool to demonstrate the concept of encryption, offering minimal security.

Vigenère Cipher extends the Caesar Cipher by using a keyword to vary the shift applied to each character. This makes it more resistant to simple frequency analysis and brute-force attacks, though long or reused keys can compromise its security. It demonstrates the principle of polyalphabetic substitution.

Transposition Cipher works by rearranging the letters in the plaintext based on a specific algorithm or key. Unlike substitution ciphers, it does not alter the actual letters, just their positions. It is useful for showing how security can be improved through permutation rather than substitution.

Uses

Caesar cipher is generally used in educational contexts, introductory lessons and simple puzzles in cryptography to explain encryption concepts.

Vigenère Cipher is used in classical encryption demonstrations, cryptography education, and historical military communications. It also appears in cryptographic challenges and ciphers in recreational settings.

Transposition Cipher finds use in cryptographic challenges, educational demonstrations, and as a foundation for more secure cipher systems when combined with substitution techniques (product ciphers).